8.6 Class Exercise: Transaction with UTXO's

Class Exercise. Follow the following instructions:

1. Understand your role: You have been assigned one of the following roles: sender, receiver, node, or miner.
- As a **sender**, you will be responsible for creating and broadcasting transactions.
- As a **receiver**, you will be responsible for receiving and verifying transactions.
- As a **node**, you will be responsible for validating the transactions and following the rules.
- As a **miner**, you will be responsible for verifying, adding the transactions to the *blockchain* and collecting rewards for your hard work.

2. If you are the **sender**, create a transaction: To create a transaction, follow these steps:
- Take a transaction form and fill in the following fields:
  - Input UTXO: 20BTC
  - Output UTXO: 10BTC to the receiver's address
  - Output UTXO:1 BTC to the miner's address
  - Change UTXO: 9 BTC to your address
  - Signature: Your signature simulating a private key.
- Pass the transaction form and the corresponding number of coins to the receiver.

3. If you are the **receiver**, Verify transactions: Follow these steps:
- Check the transaction form to ensure that the correct number of coins and the receiver's name or initials are written.
- Count the coins you received and compare them to the number of coins written on the transaction form.
- If the coins match, check the approval box on the UTXO chart that is shared and accessible to everyone in the class.
- If the coins do not match or you have doubts, reject the transaction and write the reason on the UTXO chart.

4. If you are a **node**, validate transactions: As a node, you are responsible for validating the transactions by checking that the transaction is valid by checking it against the rules of the protocol and the consensus mechanism.
- Verify that the sender's address is valid and that the receiver's address is valid.
- Check that the sender has enough funds to complete the transaction by verifying that the UTXO used as input in the transaction actually exist and have not been spent before by looking at the UTXO chart.
- Check that the transaction does not double-spend any coins by looking at the UTXO chart.